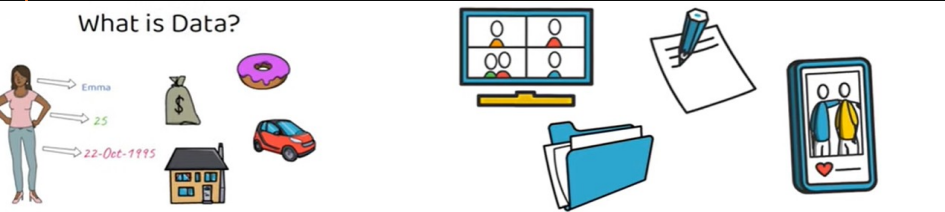


What is Data?



WHAT IS A DATABASE?

TYPES OF DATABASES?

WHAT IS A DBMS? (DATABASE MANAGEMENT SYSTEM)

WHAT IS DATA?

- A person's name, age, date of birth, address, bank balance, vehicles, school details, technology, statistics, and even the food we eat can be considered data.
- Data can also take many forms, such as images, videos, text, and documents.
- This data is stored in a database.

WHAT IS A DATABASE?

- MANY FORMS OF DATA CAN BE STORED IN A DATABASE, WHICH HELPS ORGANIZE AND MANAGE THAT DATA
- A DATABASE CAN BE THOUGHT OF AS A CONTAINER FILLED WITH DATA OR INFORMATION STORED IN A COMPUTER SYSTEM, AND MANY EVERYDAY APPLICATIONS USE DATABASES
- THE PURPOSE OF STORING DATA IS TO MAKE IT EASILY ACCESSIBLE, MODIFIABLE, PROTECTED, AND ANALYZABLE

What is Database?



Easily Accessed

Modified

Protected

Analysed

You may already be using database everyday



Building a Web Application - College Timesheet Web Application

- If a user wants to create a web application, it should be accessible through a login URL and offer options to enter staff details, timesheet details, student attendance, student marks, and so on.
- Since this is a college demo project, the student will use a personal laptop to develop it.
- 1st thing your application needs is a database to store and retrieve data. You therefore need to install a database that connects to the timesheet application, just like any other software in your application. The database is also allocated a specific location on the computer's hard drive.
- Whenever you try to save, edit, or retrieve information through your web application
- the application will connect to the database to store, modify, or extract the data.

"College Timesheet Web Application"

Daily Timesheet

Supervisor Name: Date:

Department:

Employee	Start Time	Lunch Start	Lunch End	End Time	Total Hours
Korben Dallas	9:45 AM	12:00 AM	12:30 PM	6:16 PM	8:30
					0:00
					0:00
					0:00
					0:00
Total					8:30

/Library/PostgreSQL/12

D:\Database\Oracle19c



login

localhost:8080/login

Login or Create Account

Username or Email

Password

[Forgot Password?](#)

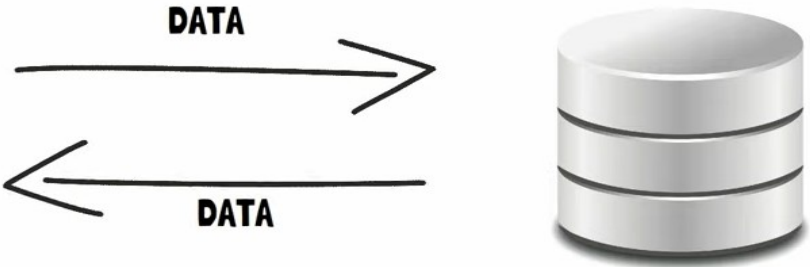
- Staff Details
- Timesheet Details
- Student Attendance
- Student Marks
- Etc...

Daily Timesheet

Supervisor Name: Date:

Department:

Employee	Start Time	Lunch Start	Lunch End	End Time	Total Hours
Korben Dallas	9:45 AM	12:00 AM	12:30 PM	6:16 PM	8:30
					0:00
					0:00
					0:00
					0:00
Total					8:30



- Since the database is installed on the computer, the data is stored in the computer's memory.
- This is a simple application to help us understand how data is stored and retrieved.
- In organizations with bigger applications, larger companies need to store huge amounts of data in large databases, often installed on large servers. That's why a server is often described as a collection of databases. All applications use databases.
- Images, videos, posts, and other information you see on social media platforms like Facebook (Meta) are actually stored in a database. Whenever a user searches for data, the specific post or user details are retrieved from the database.

Daily Timesheet

Supervisor Name: _____ Date: 14.03. 2263

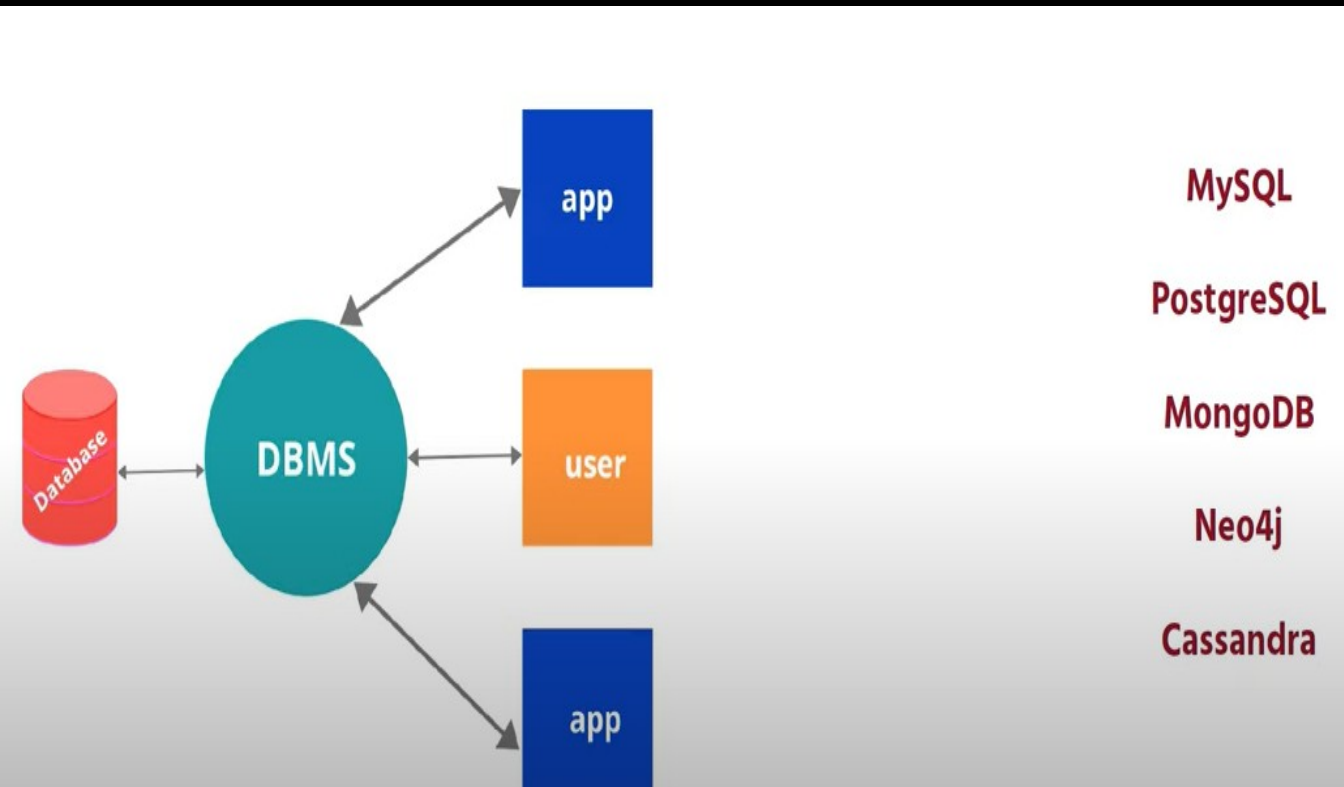
Department: _____

Employee	Start Time	Lunch Start	Lunch End	End Time	Total Hours
Korben Dallas	9:45 AM	12:00 AM	12:30 PM	6:16 PM	8:30
					0:00
					0:00
					0:00
					0:00
		Total	8:30		



facebook

DBMS $\neq$ DATABASE ?



- Is a database the same as a DBMS? Not entirely.
- Database – a container that stores data
- DBMS is software used to manage the database. You need a DBMS to interact with the database to store, modify, retrieve, and protect data.
- A DBMS is also required to create, modify, or delete a database.
- Users like you and me use a DBMS to enter commands in a specific language to interact with the database.
- Popular DBMS options include → **MySQL, PostgreSQL, Microsoft SQL Server, Oracle Database, SQLite, MongoDB, Redis, and Cassandra.**

EVOLUTION OF DATABASE

FLAT-FILE DATABASE A flat-file database is a type of database that stores data in a simple, two-dimensional table, typically as a plain text file or spreadsheet. Flat-file databases do not use multiple tables or complex relationships between tables.

HIERARCHICAL DATABASE: Organizes data in a tree-like or hierarchical structure. In this model, data is represented as a parent-child relationship, where each record (or node) has a single parent record, except for the root, and each parent can have multiple children. The structure resembles an upside-down tree or an organizational chart.

NETWORK DATABASE: Represents data using a network model. In this model, data is organized as a collection of records, and the relationships between records form a network structure. Unlike hierarchical databases, where each record has a single parent, network databases allow records to have multiple parents, enabling more flexible and complex relationships.

Evolution of Database

2020

Flat File Database



Hierarchical Database



Network Database

Relational Database

Non-Relational Database

RELATIONAL DATABASE = SQL DATABASE = STRUCTURED DATA

- In a relational database, data is stored in tables.
- Each table consists of rows and columns.
- Each column has a name and a data type.
- Each row can be treated as a record, formed by one or multiple columns.

TABLE 1					
	COLUMN 1	COLUMN 2	COLUMN 3	COLUMN 4	COLUMN 5
ROW 1	data	data	data	data	data
ROW 2	data	data	data	data	data
ROW 3	data	data	data	data	data
ROW 4	data	data	data	data	data

TABLE 2			
	COLUMN 1	COLUMN 2	COLUMN 3
ROW 1	data	data	data
ROW 2	data	data	data
ROW 3	data	data	data

TABLE 1					
	COLUMN 1	COLUMN 2	COLUMN 3	COLUMN 4	COLUMN 5
ROW 1	data	data	data	data	data
ROW 2	data	data	data	data	data
ROW 3	data	data	data	data	data
ROW 4	data	data	data	data	data

NAME	ID	DATE_OF_BIRTH	ADDRESS	GENDER	PHONE
Aaron Paul	D1	05-Jul-86	Kuala Lumpur	M	60169990102
Lara Croft	D2	01-Oct-98	Bangalore	F	9774755019
Ruth Langmore	D3	23-May-01	Singapore	F	6545459898

TABLE 3		
	COLUMN 1	COLUMN 2
ROW 1	data	data
ROW 2	data	data
ROW 3	data	data
ROW 4	data	data
ROW 5	data	data
ROW 6	data	data

TABLE 4								
	COLUMN 1	COLUMN 2	COLUMN 3	COLUMN 4	COLUMN 5	COLUMN 6	COLUMN 7	COLUMN 8
ROW 1	data	data	data	data	data	data	data	data
ROW 2	data	data	data	data	data	data	data	data
ROW 3	data	data	data	data	data	data	data	data
ROW 4	data	data	data	data	data	data	data	data
ROW 5	data	data	data	data	data	data	data	data
ROW 6	data	data	data	data	data	data	data	data
ROW 7	data	data	data	data	data	data	data	data
ROW 8	data	data	data	data	data	data	data	data
ROW 9	data	data	data	data	data	data	data	data
ROW 10	data	data	data	data	data	data	data	data

TABLE 1					
	COLUMN 1	COLUMN 2	COLUMN 3	COLUMN 4	COLUMN 5
ROW 1	data	data	data	data	data
ROW 2	data	data	data	data	data
ROW 3	data	data	data	data	data
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Ruth Langmore	D3	23-May-01	Singapore	F	6545459898
Emma Watson	D4	22-May-01			

- These tables are related through certain columns. The Employee and Manager tables are related through the manager_id column, which is present in both tables. Here, manager_id is a FOREIGN KEY.
- The Manager and Department tables are related through dept_id. The Employee and Department tables are not directly related to each other, but it is still possible to fetch data from the Employee table based on a specific condition from the Department table. This is

Department

EmpNo	EmpName	DepNo
1001	Sahil	101
1004	Kavish	102
1006	Aditya	103
1005	Atul	104

Foreign Key

Relationship

Employee

DepNo	DName	Location
101	HR	Delhi
102	Sales	Bangalore
103	Marketing Executive	Hyderabad
104	Technical Engineer	Chennai

Primary Key

Office Database

EMPLOYEE			
ID	NAME	AGE	MANAGER_ID
E101	Libinus Xavier	37	M123
E102	Gautham Bhonsle	35	M555
E103	Aravind	45	M404
E104	Shazil	28	M800
E105	Manisha Shah	34	M555

EMPLOYEE			
ID	NAME	AGE	MANAGER_ID
E101	Libinus Xavier	37	M123
E102	Gautham Bhonsle	35	M555
E103	Aravind	45	M404
E104	Shazil	28	M800
E105	Manisha Shah	34	M555

⇒

MANAGER		
ID	NAME	DEPT_ID
M123	Ravindranadh	D1011
M404	Shripad Karambelkar	D1011
M555	Meenu Dutta	D2022
M800	James Xavier	D1099
M999	Ibrahim Sheik	D1099

DEPARTMENT			
ID	NAME	DESCRIPTION	LOCATION
D1011	FINANCE	Finance Operations	Mumbai
D1099	HR	Human Resource	Bangalore
D2022	IT	Information Technology	Bangalore
D3033	ADMIN	Administrative Operations	Bangalore

↓

MANAGER		
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↓

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- Information can be retrieved from multiple tables that are related to one another. Using these table relationships, it is possible to retrieve data across different tables.
- In a relational database, using the DBMS, you can enter commands in a specific language to store, retrieve, and modify data. This language is called SQL (Structured Query Language).
- SQL is a programming language that follows a standard format for querying data across different relational databases.
- Most institutions, such as banks and insurance companies, use relational databases.
- Examples of relational databases include Oracle, Microsoft SQL Server, MySQL, and PostgreSQL.

Office Database

EMPLOYEE			
ID	NAME	AGE	MANAGER_ID
E101	Libinus Xavier	37	M123
E102	Gautham Bhonsle	35	M555
E103	Aravind	45	M404
E104	Shazil	28	M800
E105	Manisha Shah	34	M555



MANAGER		
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M123	Ravindranadh	D1011
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D1011	FINANCE	Finance Operations	Mumbai
D1099	HR	Human Resource	Bangalore
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D3033	ADMIN	Administrative Operations	Bangalore

SQL

(Structured Query Language)



NON-RELATIONAL DATABASE AKA (NoSQL DATABASE)

- Non-relational databases, often referred to as NoSQL databases, are a type of database management system that provides a mechanism for storage and retrieval of data that is modeled in ways other than the traditional relational database tables.
- These databases are designed to handle large amounts of unstructured, semi-structured, or structured data and offer flexibility and scalability that may be challenging to achieve with traditional relational databases.

Key Value Database

Document Database

Graph Database

Wide Column Database

Search Engine Database

Time Series Database

NON-RELATIONAL DATABASES (NOSQL DATABASE)

Key-Value Database:

Every piece of data is stored as a key-value pair
Once data is saved, it can be retrieved using its key.

Examples include Redis and Amazon DynamoDB .

Document Database:

Stores data in documents.

Typically uses formats like JSON or BSON (Binary JSON).

Examples include MongoDB and CouchDB.

There is no specific format to follow —
each document can have its own structure

Document Database:

MongoDB, CouchDB

Key Value Database:

KEY	VALUE
Employee_ID	EMP_1002398_C1
Status	Active
Salary	25000
Joining_Date	17-Feb-2016
Details	{ "Location": "New York", "Project": "NewStore", "Skills": ["SQL", "Python"] }

Director database	
Host Name	Gonville
Port number	1521
Datasource UserName	Director
Datasource Password	
Database ID	ORCL
Database Type	Oracle
Results database	
Host Name	Gonville
Port number	1521
Datasource UserName	Results
Datasource Password	
Database ID	ORCL

Document Database:

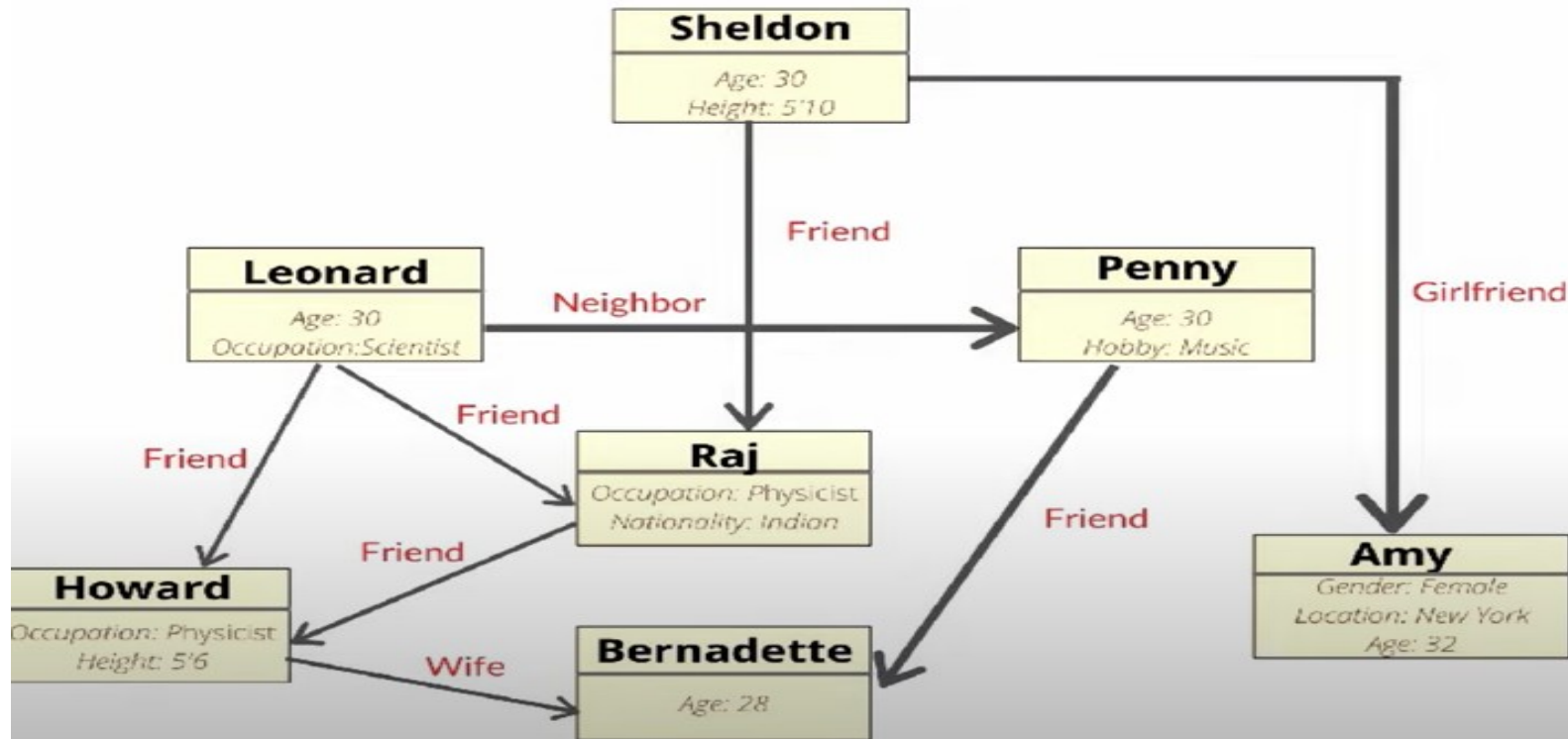
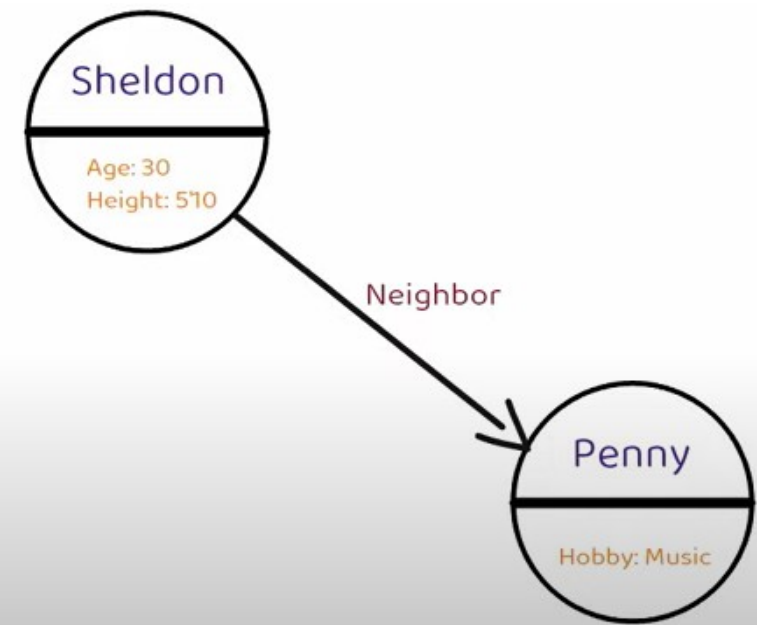
ID: iPad
{ "Type": "Tablet", "Manufacturer": "Apple", "Apps": ["Safari", "FaceTime"] }

ID: MacBook Pro
{ "Type": "Laptop", "Colour": "Space Grey", "Processor": ["Intel i5", "Quad core", "10th generation", "2.0GHz", "Turbo": ["Turbo Boost upto 3.8+GHz", "6MB shared L3 cache",]], "price": "RM 8,799", "Manufacturer": "Apple", "Display": ["13.3-inch", "2560x1600 resolution", "227 pixels per inch", "500 nits brightness"], "Memory": "16 GB LPDDR4X", "Graphics": "Intel Iris Plus Graphics" }

ID: iPhone
{ "Type": "Phone", "Colour": "Green", "Camera": ["Dual 12MP Ultra Wide", "Wide: f/1.8 aperture", "2x optical zoom", "Night mode"], "price": "RM 3,500", "Capacity": "64 GB", "Chip": "A13 Bionic chip" }

GRAPH DATABASE

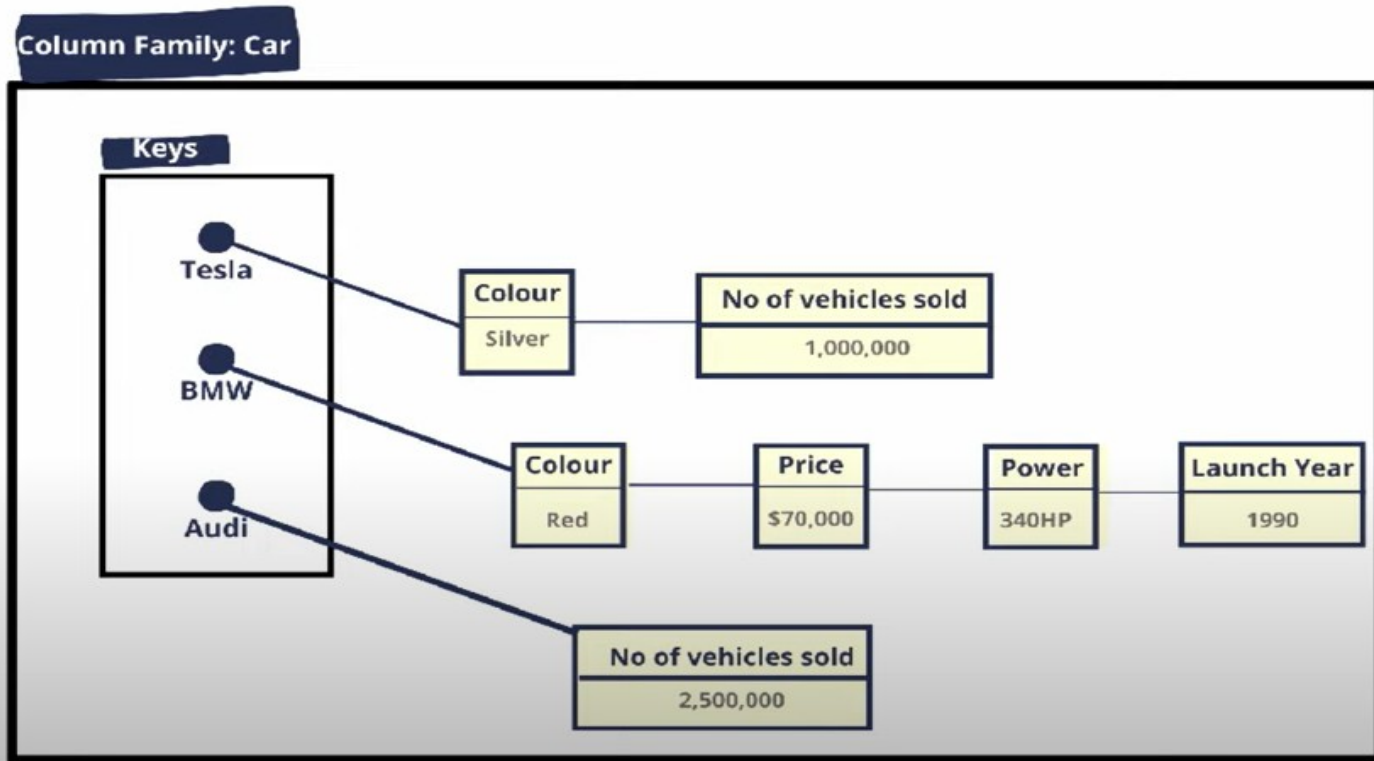
- Designed for storing and querying graph-like data structures.
- Examples include Neo4j and Amazon Neptune.
- Forms relationships using nodes, edges, and properties.
- Data is represented as nodes with multiple properties. Between nodes, edges define the type of connection.
- Useful for finding specific patterns, such as money laundering, fraud detection.



WIDE-COLUMN DATABASE or COLUMN-FAMILY STORES:

- Stores data in columns rather than rows, suitable for large amounts of data with varying attributes. Examples: Apache Cassandra and HBase.
- Wide-column databases store data using rows and columns, but they don't use tables. Instead of tables, they use a structure called a column family.
- A column family contains rows of data, where each row can have its own structure or schema.

Wide Column Database



Cassandra, HBase etc.

What is Database?

- A database is an organized collection of data that can be modified, retrieved, or updated.
- The data, stored in the database, is in the row and column format, which is called a **table**.
- Every website, which needs us to sign up, uses a database.
- There is **no internet** without databases.



Hierarchical databases

- A hierarchical database is a data model in which data is stored in the form organized into a **tree-like** structure, or parent-child structure, in which one node can have many child nodes connected through links. Example - **IBM Information Management System (IMS)**

Types of Databases

Object-oriented databases

- Object-oriented databases (OODB) are databases that represent data in the form of **objects** and **classes**. In object-oriented terminology, an object is a real-world entity, and a class is a collection of objects. Example - **PostgreSQL**

Relational databases

- A relational database is a collection of data items with pre-defined **relationships** between them. These items are organized as a set of **tables** with **columns** and **rows**. Example - **MySQL**

Distributed databases

- A distributed database (DDB) is an integrated collection of databases that is **physically distributed** across sites in a computer network. Example - **Cassandra**

What Is a DBMS?

- Database Management System (DBMS) is a **software** for storing and retrieving users' data while considering appropriate security measures.
- It consists of a group of programs which **manipulate** the database.
- The DBMS accepts the request for data from an application and instructs the operating system to provide the specific data.
- In large systems, a DBMS helps users and other third-party software to store and retrieve data.
- DBMS allows users to create their own databases as per their requirement. It provides an interface between the data and the software application.

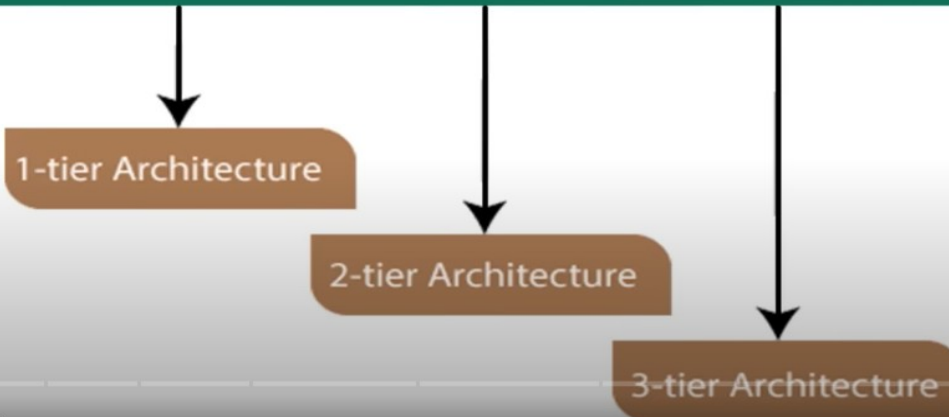
Applications Of DBMS



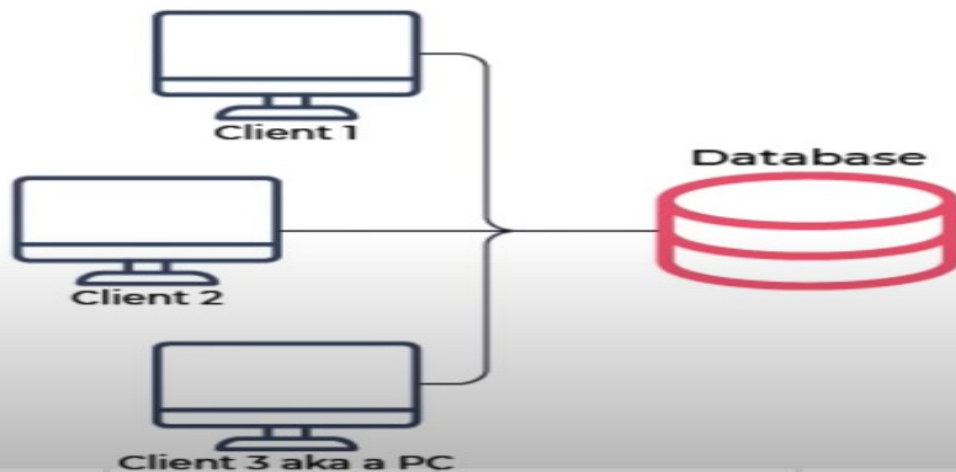
HR Management	For information about employees, salaries, payroll, generation of paychecks, etc.
---------------	---

Banking	For customer information, account activities, payments, deposits, loans, etc.
Airlines	For reservations and schedule information.
Universities	For student information, course registrations, colleges and grades.
Telecommunication	It helps to keep call records, monthly bills, maintaining balances, etc.
Finance	For storing information about stock, sales, and purchases of financial instruments like stocks and bonds.
Sales	Use for storing customer, product & sales information.
Manufacturing	It is used for the management of supply chain and for tracking

Database Architecture in DBMS



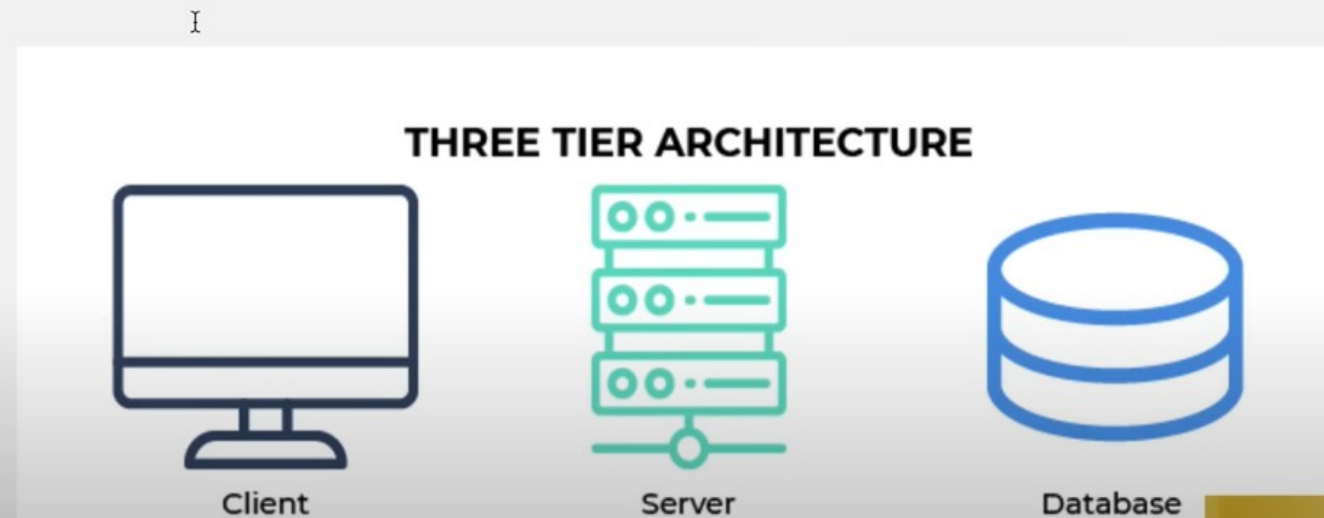
2-Tier Architecture : A 2 Tier Architecture in DBMS is a Database architecture where the presentation layer runs on a client (PC, Mobile, Tablet, etc.), and data is stored on a server called the second tier. Two tier architecture provides added security to the DBMS as it is not exposed to the end-user directly. It also provides direct and faster communication.



- **1-Tier Architecture :** 1 Tier Architecture in DBMS is the simplest architecture of Database in which the client, server, and Database all reside on the same machine. A simple one tier architecture example would be anytime you install a Database in your system and access it to practice SQL queries. But such architecture is rarely used in production.



3-Tier Architecture : A 3 Tier Architecture in DBMS is the most popular client server architecture in DBMS in which the development and maintenance of functional procedure logic, data access, data storage, and user interface is done independently as separate modules. Three Tier architecture contains a presentation layer, an application layer, and a database server.



What is my sql? ||
How to Install
MySQL

How to Create a
Table in MySQL?

MySQL Add Data
Into Tables using the
INSERT Query

MySQL Select Query
with Where Clause

MySQL Table
Constraints Tutorial
with Example

MySQL Commands:
And, OR & NOT
Operator

MySQL IN Operator
on Database Table

MySQL LIKE
Operator and
Wildcard

MySQL Between and
Not Between
Operators

MySQL Order By and
Distinct

MySQL IS NULL and
IS NOT NULL
Operators

MySQL Limit and
Offset Complete
Tutorial

MySQL Aggregate
Functions: SUM,
MIN, MAX & AVG

MySQL UPDATE
Statement

MySQL DELETE
Query

Primary Key and
Foreign Key using
MySQL Command

MySQL Data
Transaction

Primary Key and
Foreign Key in
MySQL

MySQL INNER JOIN
Command

What is the
Difference Between
Left Join & Right Join
in MySQL?

Cross Join in MySQL

How to Join Multiple
Tables using MySQL
Command?

What are MySQL
GROUP BY &
HAVING

Explanation of
MySQL SubQuery
with EXISTS & NOT
Exists

What is MySQL

MySQL is a relational **database management system** (**RDBMS**) developed by Oracle that is based on structured query language (**SQL**).

What is Database?

A **database** is an organized collection of structured information, or data.

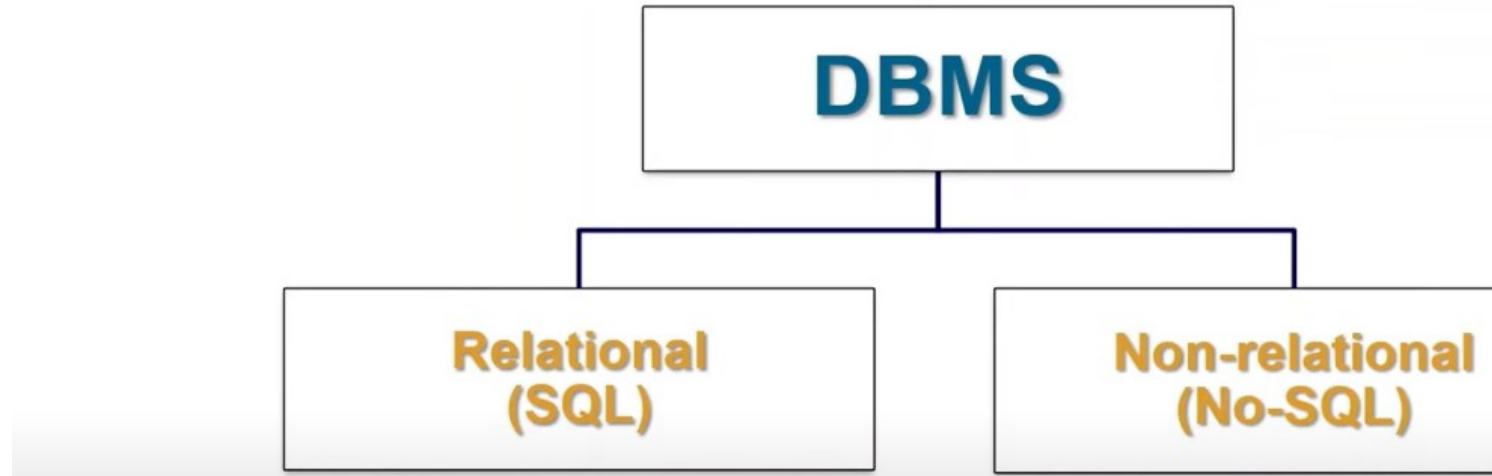
What is DBMS?

**CRUD OPERATIONS
PERFORMED BY A DBMS**

Database Management Systems (**DBMS**) are software systems used to store, retrieve, and run queries on data.

C → Create
R → Read
U → Update
D → Delete

Types of DBMS?

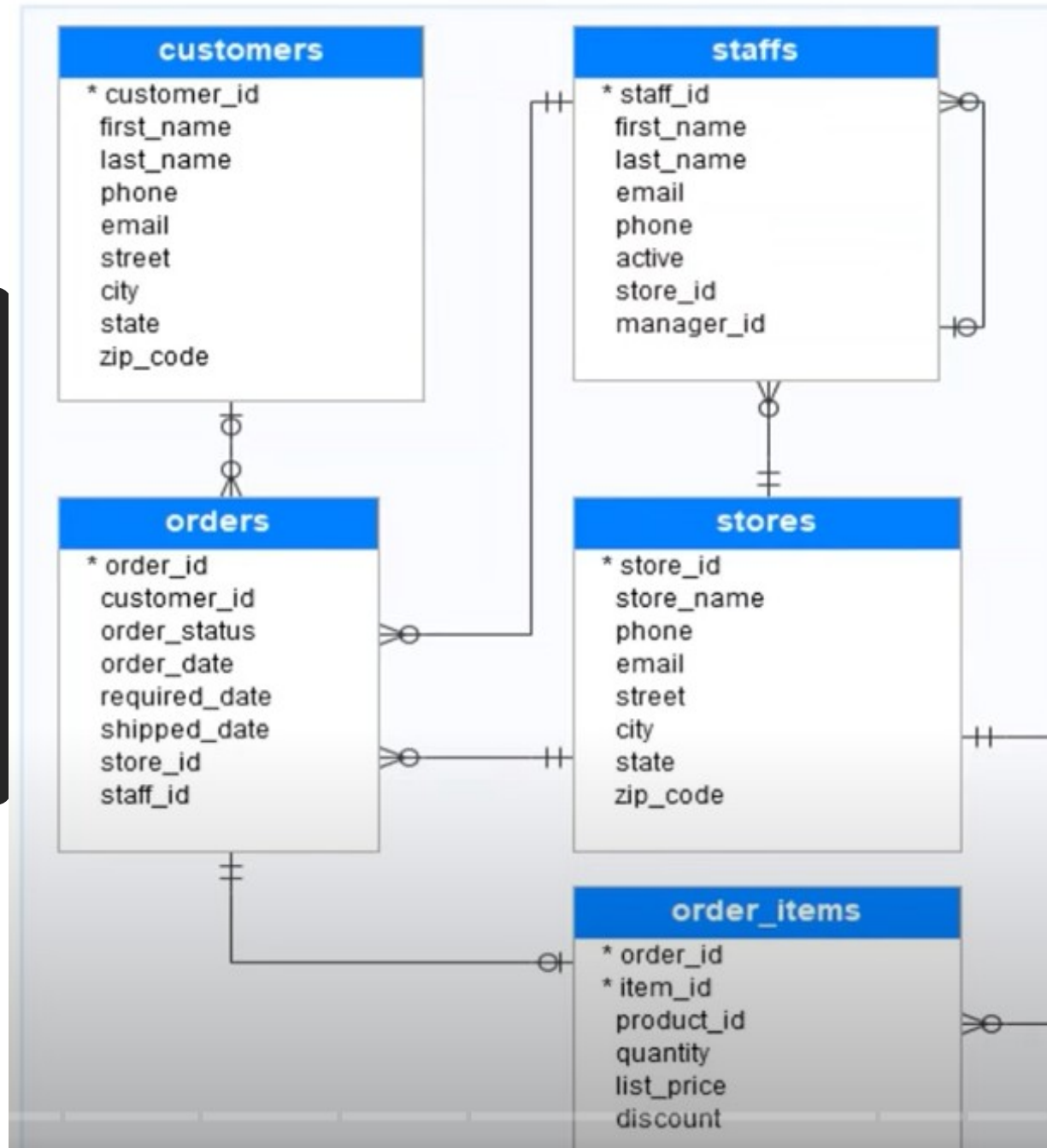


What is **Relational** database?

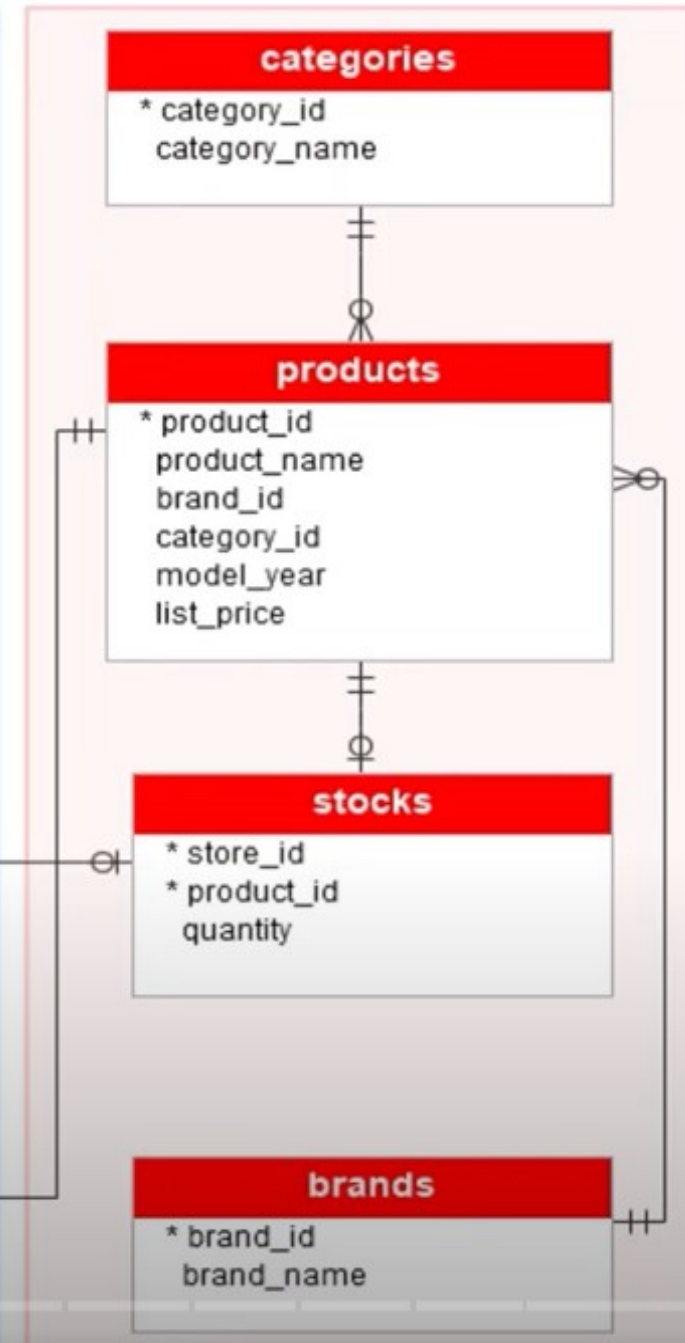
A relational database is a collection of data items with pre-defined relationships between them, stored in the form of table, rows and columns

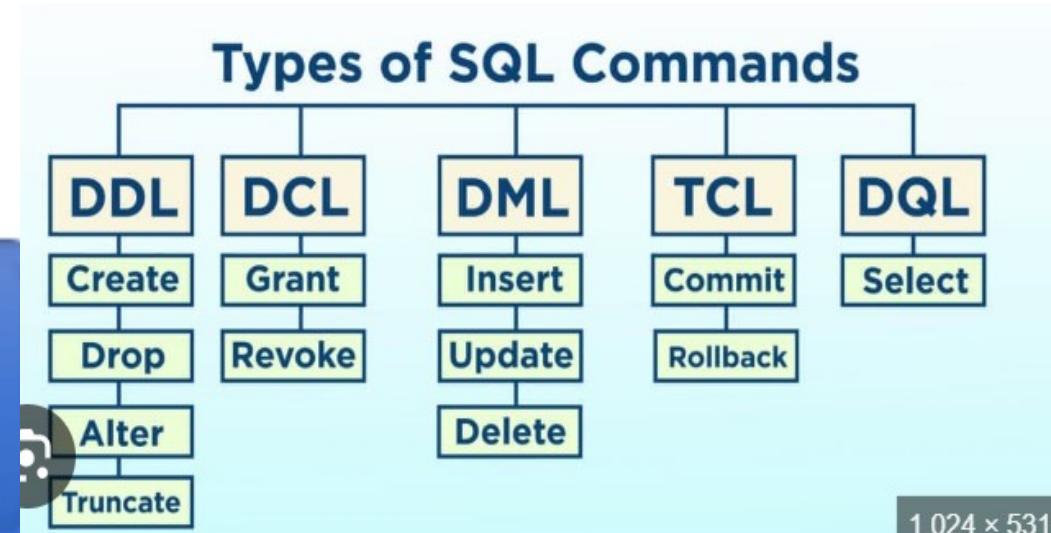
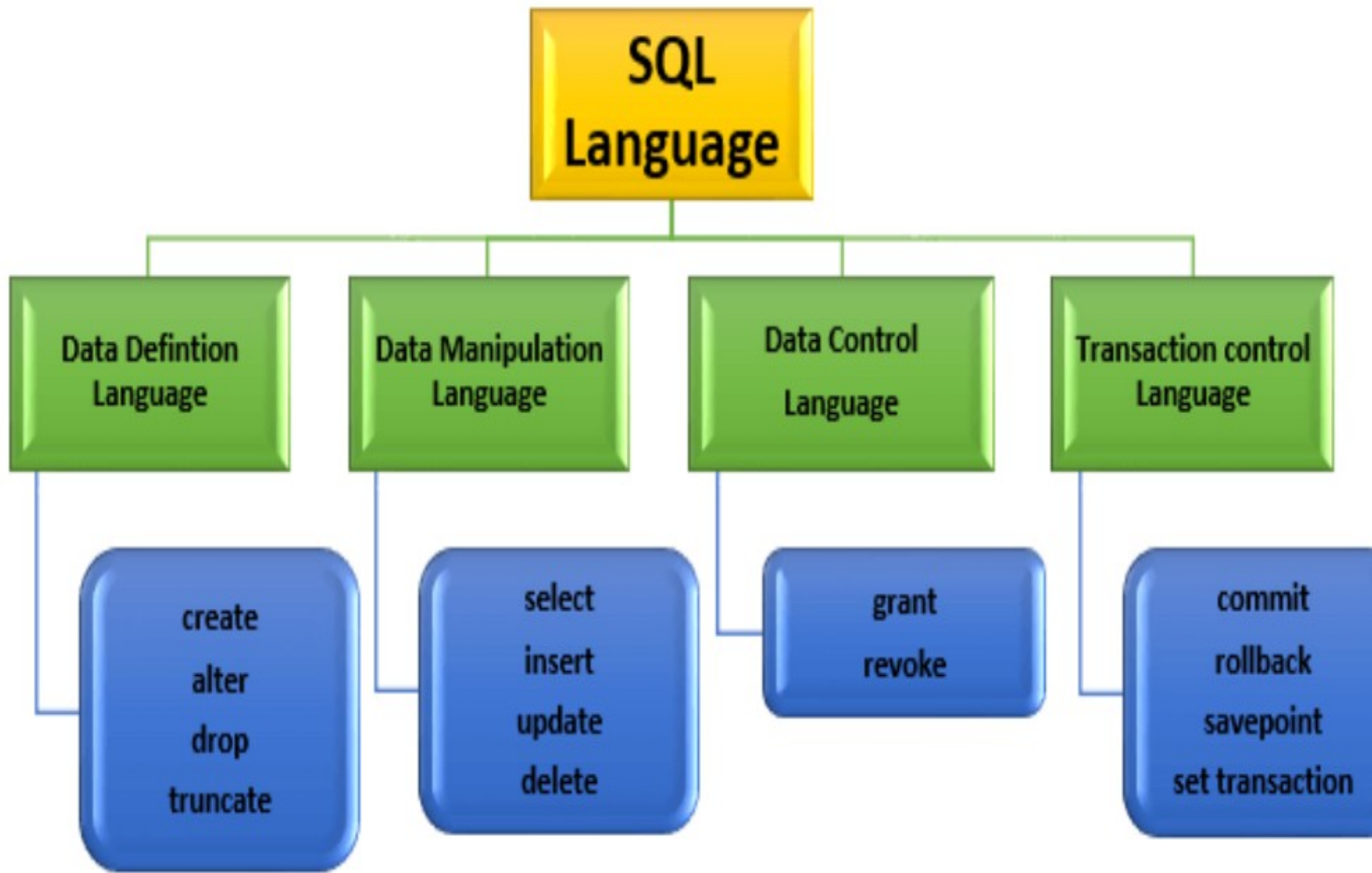
RELATIONAL DATABASE

Sales



Production

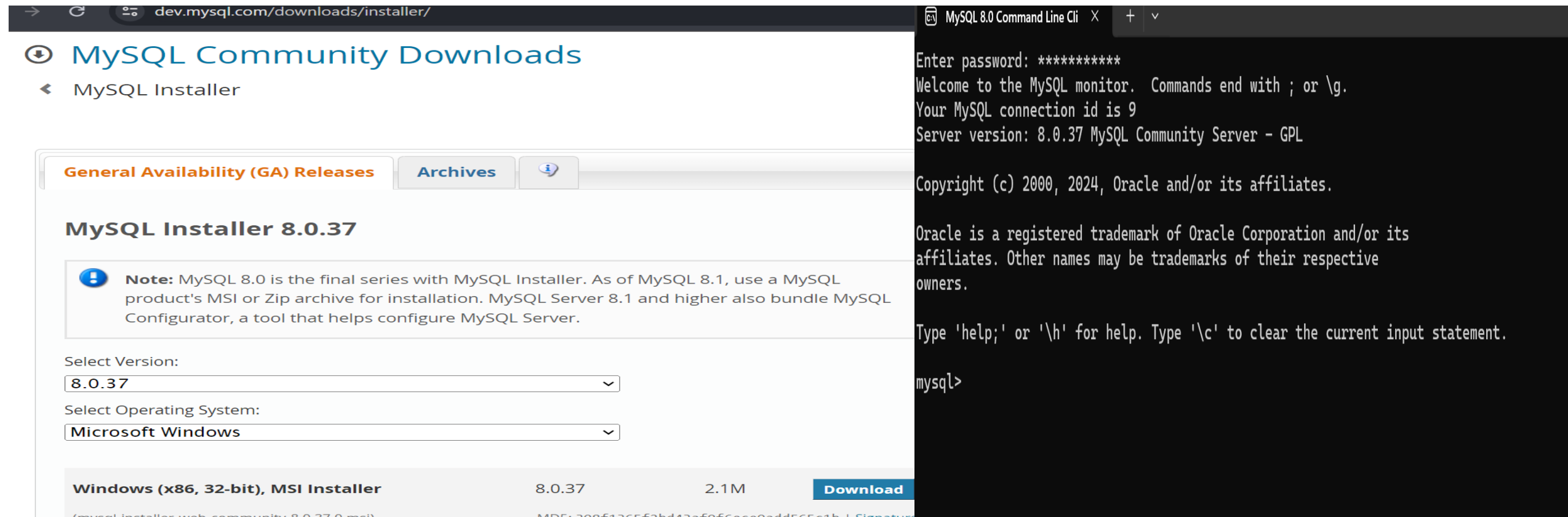




COMPLETE SQL PRACTICAL

MySQL is one of the most widely used databases in the world. Here are the steps to download and install MySQL: ➔

1. Search for MySQL download on Google, then click on MySQL Installer, or use the link below.



The image shows two side-by-side screenshots. The left screenshot is from the MySQL Community Downloads website, specifically the 'MySQL Installer 8.0.37' page. It features a navigation bar with 'General Availability (GA) Releases', 'Archives', and an information icon. A note states: 'Note: MySQL 8.0 is the final series with MySQL Installer. As of MySQL 8.1, use a MySQL product's MSI or Zip archive for installation. MySQL Server 8.1 and higher also bundle MySQL Configurator, a tool that helps configure MySQL Server.' Below the note, there are two dropdown menus: 'Select Version:' with '8.0.37' selected, and 'Select Operating System:' with 'Microsoft Windows' selected. At the bottom, a table lists the download options:

Operating System	Version	Size	Action
Windows (x86, 32-bit), MSI Installer	8.0.37	2.1M	Download

The right screenshot is a terminal window titled 'MySQL 8.0 Command Line Cli'. It shows the following text:

```
Enter password: *****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 9
Server version: 8.0.37 MySQL Community Server - GPL

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql>
```

2. <https://dev.mysql.com/downloads/installer/> — click “No thanks, just start my download.”

3. Click “Next,” set the root password, and finish the installation process.

4. Open MySQL 8.0 Command Line Client and enter your root password when prompted.